

Purpose

Set up a working PIC16F883 development environment, establish a reusable PIC-AS project workflow, & build progressively more complex digital I/O applications.

This is the first PIC assembly Lab so parts 1 & 2 are more guided. Parts 3 & 4 require independent design & debugging

Standards & reusable resources

Use these throughout the assignment [Github links in blue](#)

- RCET 3375 Lab standard
- Starting a PIC-AS Project
- PIC16F883 4MHz crystal Oscillator
- RCET PIC-AS style Guide
- PIC16F883 Data sheet

Equipment & Components

- Computer with MPLab X IDE 6.20 & Pic-as 3.10
- PICKit 3
- PIC16F883
- 4MHz crystal & required capacitors
- Oscilloscope
- Logic analyzer Optional
- DMM
- DLG7137 display
- 8 Position Dip switch assembly
- 4x4 matrix Keypad
- Resistors & other interference components as needed
- Lab book

Part 1 - First PIC-AS ProjectGoal

create a known-good PIC16F883 project that can be used as a starting point for later assignments

Before Lab

Follow Starting a PIC-AS Project & the RCET 3375 Lab Standard

Prepare:

- One git repository for this assignment with directory structure.
- ~~MPLAB~~ MPLab XIDE 6.20 & Pic-as 3.10
- A standalone PIC16F883 Project under Source/PIC/
- PICKit 3 Project Configuration
- deliberate Config-bit settings
- reset & interrupt PSECT Placement at the correct addresses
- a course-standard main.S starter file
- An external Oscillator with selected load capacitors

Document all SFRs, Configuration decisions, IC pins, & loading checks

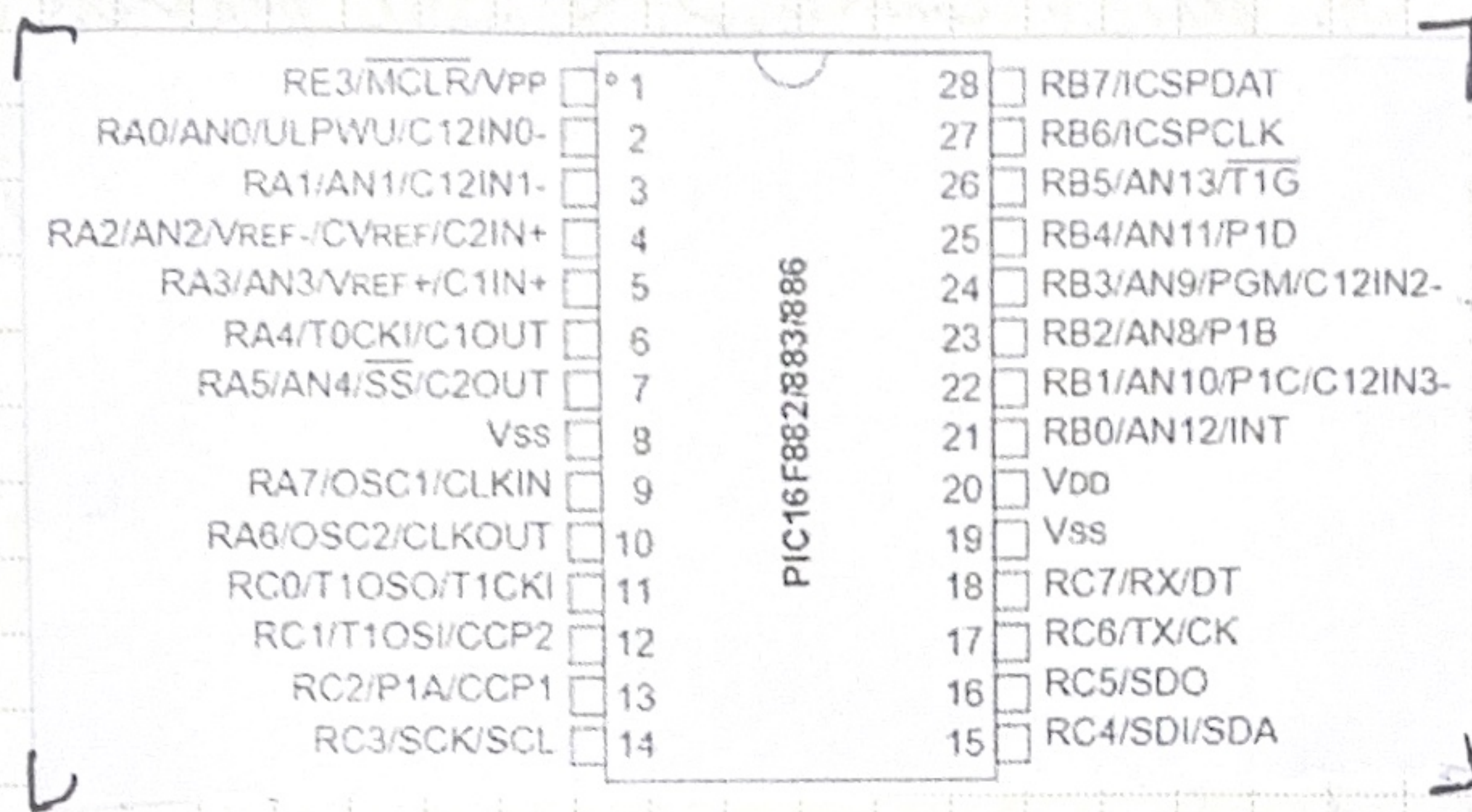
4MHz crystal
PIC oscillator

Experiment 2 PIC16F883 I/O Familiarization

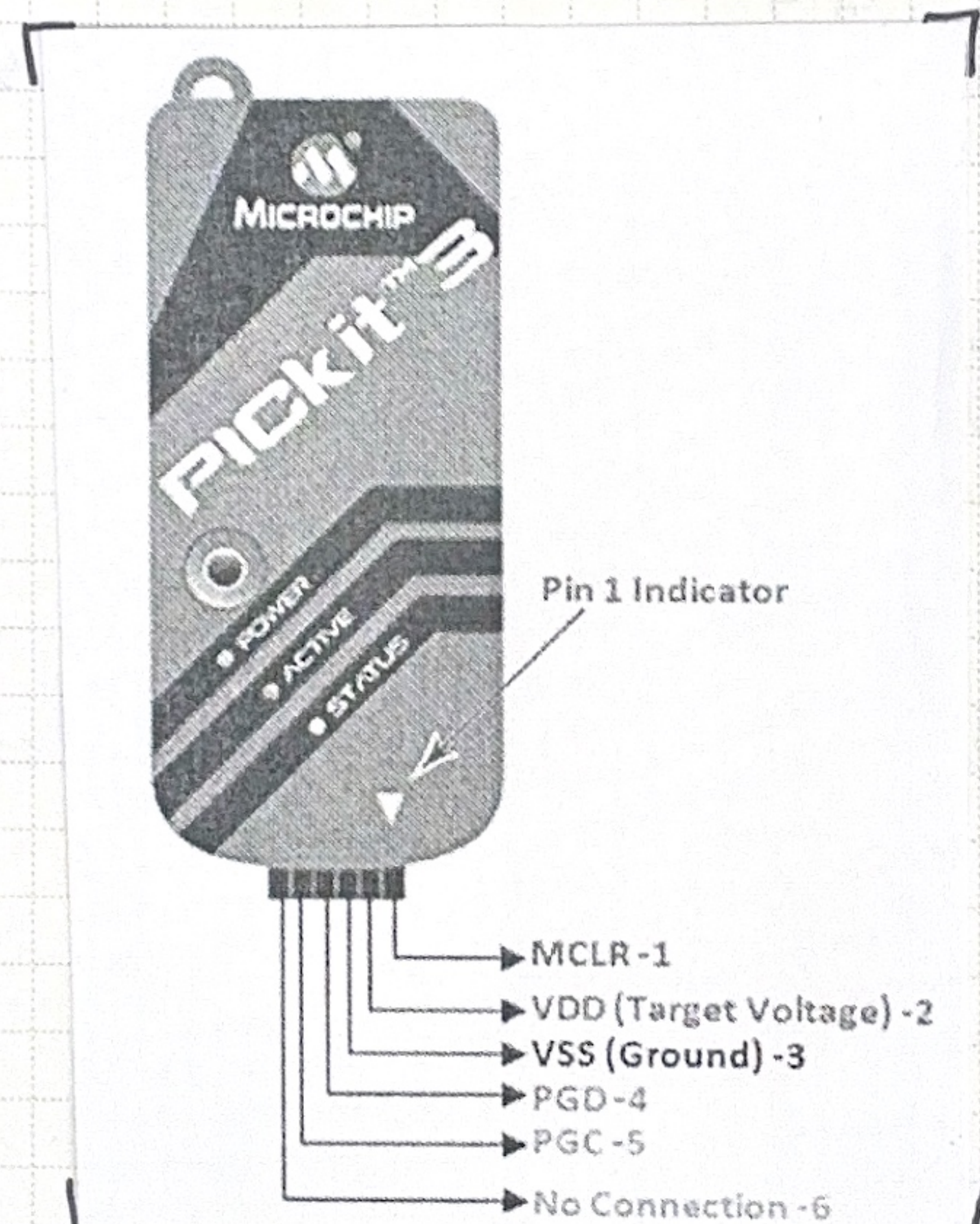
ART
9/1/26
Applying power

In The Lab

1. Inspect power, reset, oscillator, & PICkit Connections before

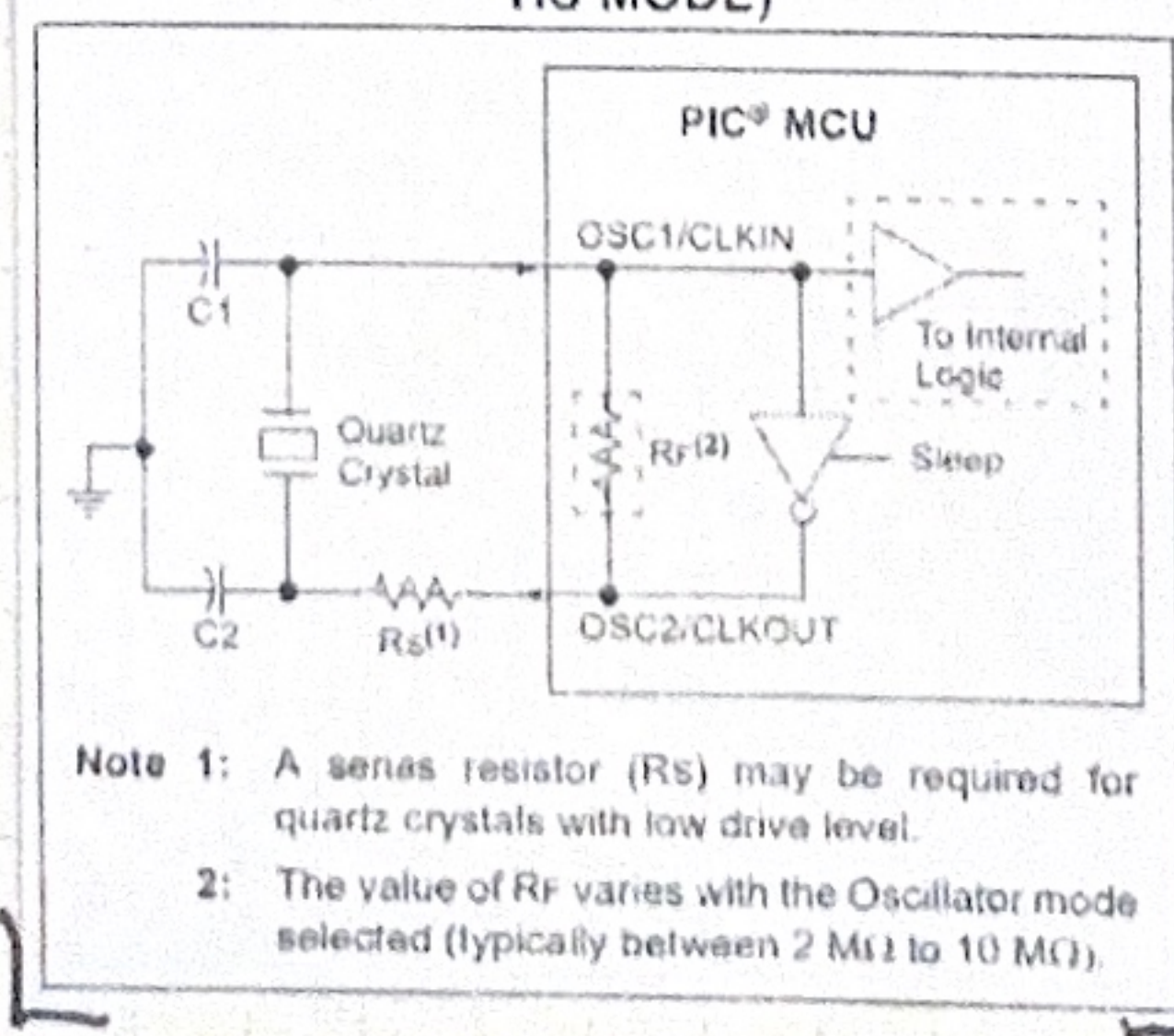


PIC16F883 Pinout Connections



PICkit Pinout Connections

FIGURE 4-3: QUARTZ CRYSTAL OPERATION (LP, XT OR HS MODE)



Crystal Oscillator Build for XT Mode

Table 2-4: Typical Capacitor Selection for Crystal Oscillator

Mode	Freq	C1 ⁽¹⁾	C2 ⁽¹⁾
LP	32 kHz	68 - 100 pF	68 - 100 pF
	200 kHz	15 - 30 pF	15 - 30 pF
XT	100 kHz	68 - 150 pF	150 - 200 pF
	2 MHz	15 - 30 pF	15 - 30 pF
	4 MHz	15 - 30 pF	15 - 30 pF
HS	8 MHz	15 - 30 pF	15 - 30 pF
	10 MHz	15 - 30 pF	15 - 30 pF
	20 MHz	15 - 30 pF	15 - 30 pF

Crystals used:

32.768 kHz	Epson C-001R32-768K-A	± 20 PPM
100 kHz	Epson C-2 100.00 KC-P	± 20 PPM
200 kHz	STD XTL 200.000 kHz	± 20 PPM
2.0 MHz	ECS ECS-20-S-2	± 50 PPM
4.0 MHz	ECS ECS-40-S-4	± 50 PPM
10.0 MHz	ECS ECS-100-S-4	± 50 PPM
20.0 MHz	ECS ECS-200-S-4	± 50 PPM

Table 1-1 Capacitor Selection for crystal

The config code written by MPLab set the oscillation mode to XT which is the PIC's medium gain mode best for external resonators/crystal oscillators. This default setup means we don't need to initiate any code to connect the 4MHz crystal oscillator.

2. Verify the MPLab recognizes the PICKit3 & target device

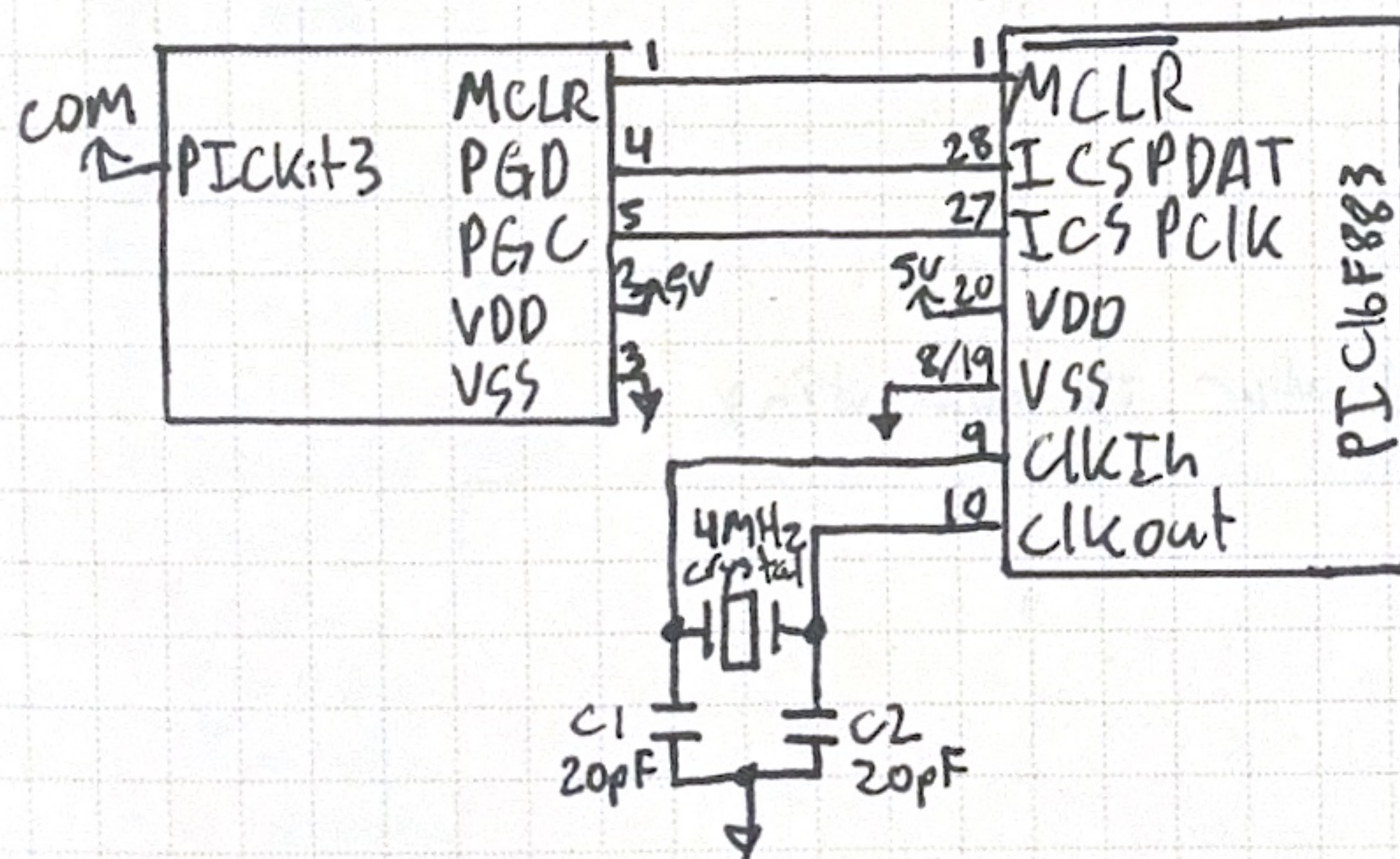


Figure 1 PICKit programming Setup configuration

*PICKit 3 users Guide states the Target device (PIC16F883) should be powered a 1.8-5V so VDD will be 5V

Experiment 2 ~~PIC16F883~~ PIC16F883 I/O Familiarization
3. Build the starter project without assembler or linker errors. ART 9/1/26

Errors can be corrected by inputting this string into the
Pic-as Global options Additional Settings: ~~-Wl, -PresetVect=0000~~
~~-Wl, -presetVect=0000h, -pcode=0008h~~
Find this in the project properties.

PIC16F883 Configuration bit Settings

Assembly source line config statements

```
; CONFIG1
CONFIG FOSC = XT           ; Oscillator Selection bits (XT oscillator: Crystal/resonator on RA6/OSC2/CLKOUT and RA7/OSC1/CLKIN)
CONFIG WDTE = OFF          ; Watchdog Timer Enable bit (WDT disabled and can be enabled by SWDTEN bit of the WDTCON register)
CONFIG PWRTE = OFF         ; Power-up Timer Enable bit (PWRT disabled)
CONFIG MCLRE = ON          ; RE3/MCLR pin function select bit (PE3/MCLR pin function is MCLR)
CONFIG CP = OFF            ; Code Protection bit (Program memory code protection is disabled)
CONFIG CPD = OFF           ; Data Code Protection bit (Data memory code protection is disabled)
CONFIG BOREN = OFF         ; Brown Out Reset Selection bits (BOR disabled)
CONFIG IESO = OFF          ; Internal External Switchover bit (Internal/External Switchover mode is disabled)
CONFIG FCMEN = OFF         ; Fail-Safe Clock Monitor Enabled bit (Fail-Safe Clock Monitor is disabled)
CONFIG LVP = OFF           ; Low Voltage Programming Enable bit (RB3 pin has digital I/O, HV on MCLR must be used for programming)

; CONFIG2
CONFIG BOR4V = BOR40V      ; Brown-out Reset Selection bit (Brown-out Reset set to 4.0V)
CONFIG WRT = OFF           ; Flash Program Memory Self Write Enable bits (Write protection off)
```

// config statements should precede project file includes.

#include <xc16f.h>

4. Program the PIC16F883

Target voltage detected
Target device PIC16F883 found.
Device Revision ID = 0

Device Erased...

Programming...

The following memory area(s) will be programmed:
program memory: start address = 0x0, end address = 0xb
configuration memory

Programming/Verify complete

MPLab Output checks

5. Verify the 4MHz Oscillator is operating

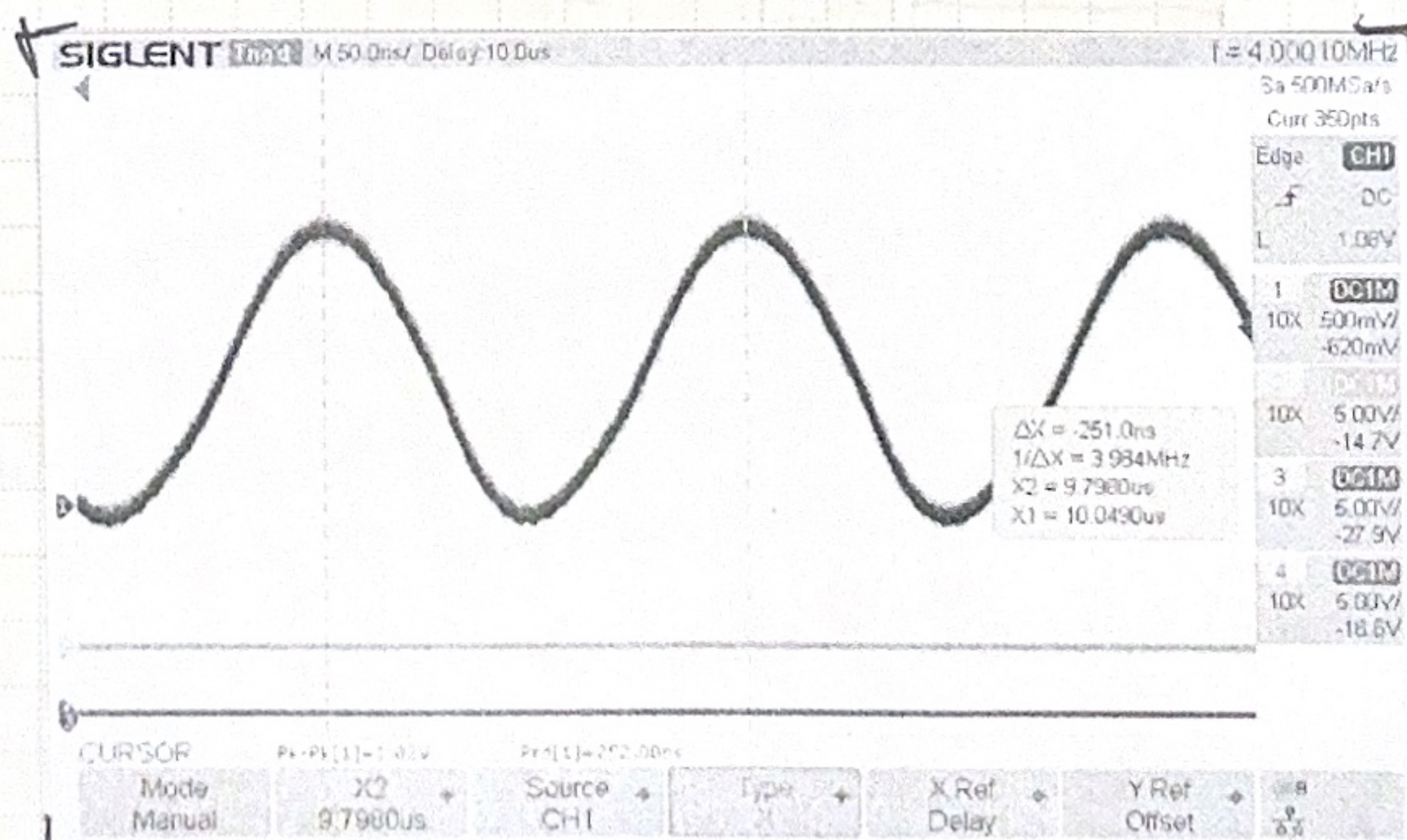


Figure 2 Measured CLKout @ 4MHz

6. Verify the reset & interrupt PSECT locations from build output 9/12/26

Program Memory x				
Line	Address	Opcode	Label	DisAssy
1	000	0000		NOP
2	001	0000		NOP
3	002	0000		NOP
4	003	2808		GOTO 0x8
5	004	3FFF		ADDLW 0xFF
6	005	3FFF		ADDLW 0xFF
7	006	3FFF		ADDLW 0xFF
8	007	3FFF		ADDLW 0xFF

Table 1-2
MPLab Program Memory

7. Commit the known good project & push it to Git hub.

8. Verify Git status is clean

```

kidhed@AshtonCom MINGW64 ~/OneDrive/Attachments/College/Semester5/Lab2 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

```

Git
Bash Clean
Verification

<https://github.com/Ashton-Tolman/Semester-5-Lab-2.git>

← Repository Link

* following the Repository Link should show the complete code with the program memory & PSECT Vector info. The build is in PIC.pl & is successful. Git commit info is included

Part 2 - PORTB Output Counter

Goal

Configure PortB as a digital output & continuously increment it. Connect the SFR configuration & assembly program to observable hardware behavior.

Before Lab

Use the PIC16F883 datasheet to determine all configuration required for PortB digital output

Prepare;

- Complete SFR records or earlier Lab book page references.
- Updated schematic
- Loading Calculations for every pin/component
- Program flowchart
- PIC-AS source
- a prediction of the portB output sequences & RBO behavior

Experiment 2 PIC16F883 I/O Familiarization (Full SFR Lists on pgs 309-312)

ART
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Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Register on Page
ANSELH	---	---	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8	49
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	122
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	---	---	T1GSS	C2SYNC	92
IOCB	IOCB7	IOCB6	IOCB5	IOCB4	IOCB3	IOCB2	IOCB1	IOCB0	50
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	32
OPTION_REG	RBPUR	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	31
PORTB	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	49
TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	49
WPUB	WPUB7	WPUB6	WPUB5	WPUB4	WPUB3	WPUB2	WPUB1	WPUB0	50

Table 2-1 SFR's Associated with port B

PSECT resetVect, class=CODE, delta=2

ResetVector:

goto Setup

; Main code section starts at 0008h. User code goes here
PSECT code, class=CODE, delta=2

Setup:

; add device setup here

```
BSF 0x03, 6 ;
BSF 0x03, 5 ; Set bank 3 active
CLRF 0x89 ; ANSELH analog/digital select portB pins as digital

BSF 0x03, 6 ;
BCF 0x03, 5 ; Set bank 2 active
BSF 0x09, 1 ; Turn T1G-not function off
BCF 0x0B, 0 ; Turn INTCON RBIF flag off, other bits are default off

BCF 0x03, 6 ;
BSF 0x03, 5 ; Set bank 1 active
CLRF 0x06 ; TrisB sets all portB pins as outputs

BCF 0x03, 6 ;
BCF 0x03, 5 ; Set bank 0 active
CLRF 0x07 ; CCP1CON capture/compare/PWM off

; OPTION_REG RBPUR-not is default high meaning disabled
; WPUB weak pull ups default disabled
; IOCB portB interrupt-on-change disabled default
```

goto Main

Main:

// main goes here

```
INCF 0x06, 1 ; Bank 0 is already active, increment portB on a loop
goto Main
```

PIC16F883 Config settings

Reset Vector at 0x000h

Jump to Setup

Setup Phase

Jump to Main

Main Loop

Increment PortB

Infinite Loop?

Code Flowchart
Figure 2-1

MPLab SFR code & instructions
Based of SFR's Associated with Port B
Figure 2-2

The main code instruction will increment in binary the 8-bits on port B meaning RB0, which follows the 0 bit of port B, will be pulsing at every point the instruction is applied which should be at every 3µs. This is because the 4MHz frequency must clock 4 times for one instruction cycle. So one instruction happens every 1µs with 3 instructions, 1 for increment, 2 for goto Main, it means RB0 clocks every 3µs.

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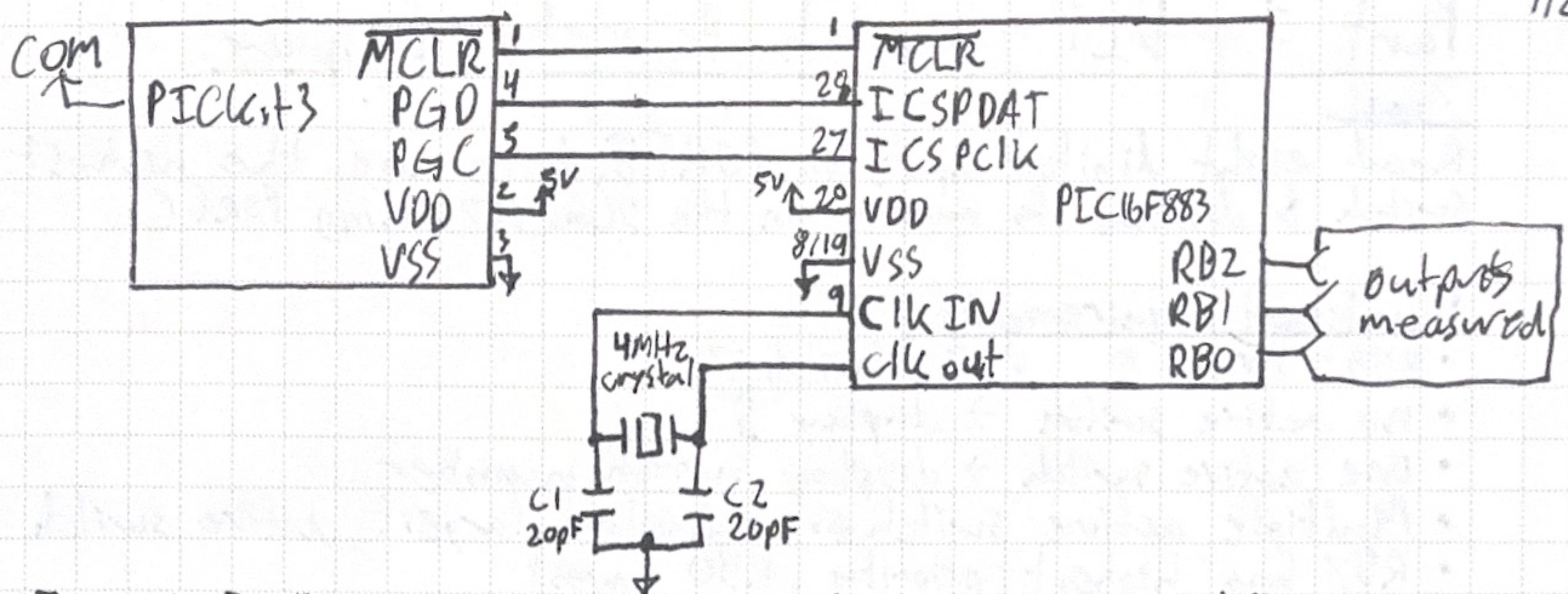


Figure 2-3 Programming Circuit with measured PortB bits
Crystal CLK Loading from data sheet (pg 145)

In the Lab

1. Build & program the PIC
2. Verify that PortB continuously increments
3. Observe enough PortB lines to confirm binary counting
4. Capture RBO with the Oscilloscope
5. Measure High pulse width, Low pulse space, Period & Frequency
6. Compare the measurements with your prediction

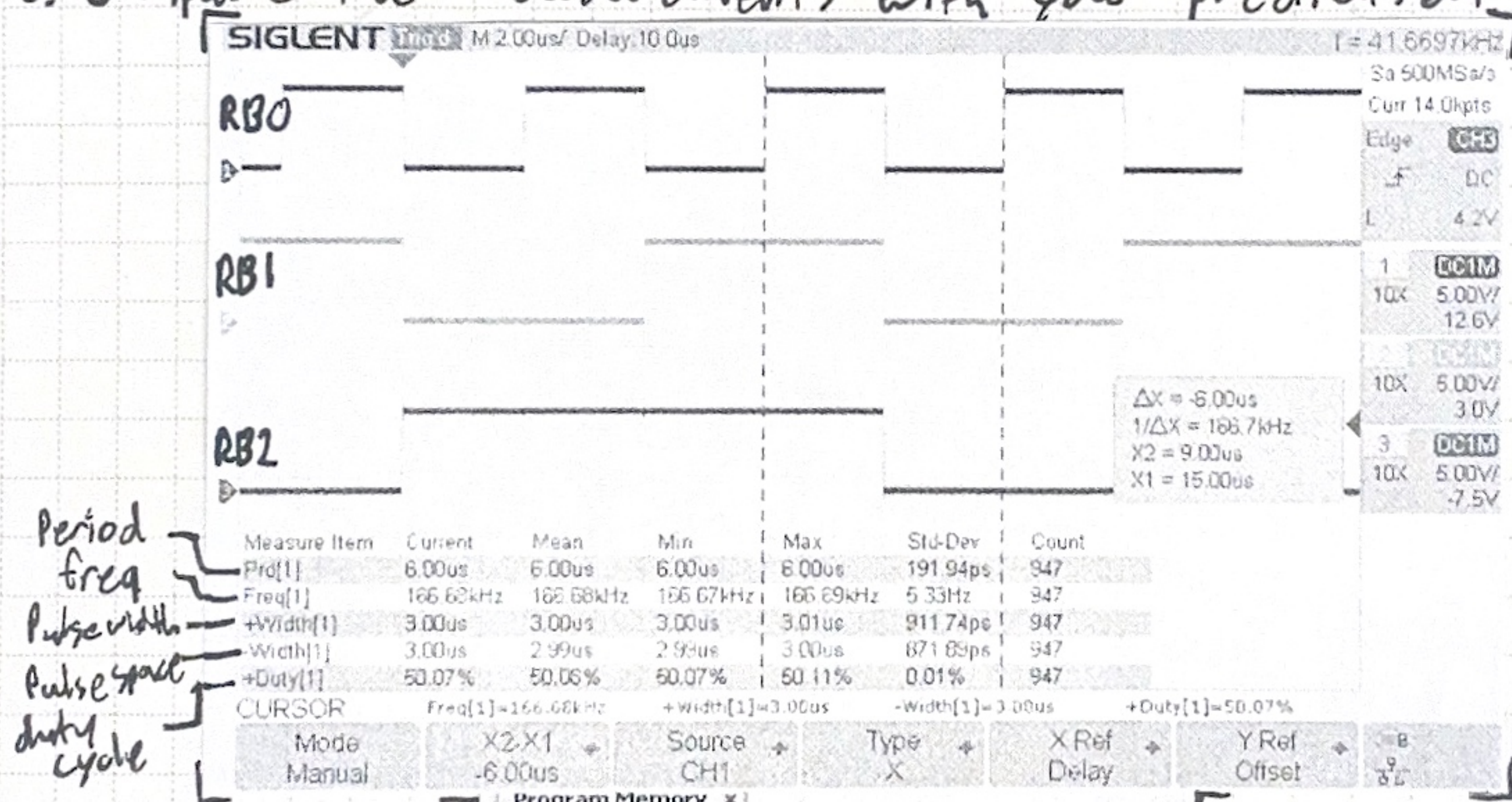


Figure 2-4
measured
values
from PortB

Line	Address	Opcode	Label	DisAssy
1	000	2808		GOTO 0x8
2	001	3FFF		ADDLW 0xFF
3	002	3FFF		ADDLW 0xFF
4	003	3FFF		ADDLW 0xFF
5	004	3FFF		ADDLW 0xFF
6	005	3FFF		ADDLW 0xFF
7	006	3FFF		ADDLW 0xFF
8	007	3FFF		ADDLW 0xFF
9	008	1703		BSF STATUS, 0x6
10	009	1683		BSF STATUS, 0x5
11	00A	0189		CLRF PORTE
12	00B	1703		BSF STATUS, 0x6
13	00C	1283		BCF STATUS, 0x5
14	00D	1489		BSF PORTE, 0x1
15	00E	100B		BCF INTCON, 0x0
16	00F	1303		BCF STATUS, 0x6
17	010	1683		BSF STATUS, 0x5
18	011	0186		CLRF PORTB
19	012	1303		BCF STATUS, 0x6
20	013	1283		BCF STATUS, 0x5
21	014	0187		CLRF PORTC
22	015	2816		GOTO 0x16
23	016	0A86		INCF PORTB, F
24	017	2816		GOTO 0x16

Table 2-2 Program memory

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Part 3 - DIP-Switch Priority Display

Goal

Read eight digital inputs on PORTB, determine the highest active switch, & display its number on the DLG2137 using PORTC

Functional requirements

- RB0 = switch 0 RB7 = Switch 7
- no active switch → display \$
- One active switch → display switch number
- Multiple active switches → display largest active switch number
- RB7 has highest priority, RB0 lowest
- Operation is continuous

Before Lab

Prepare:

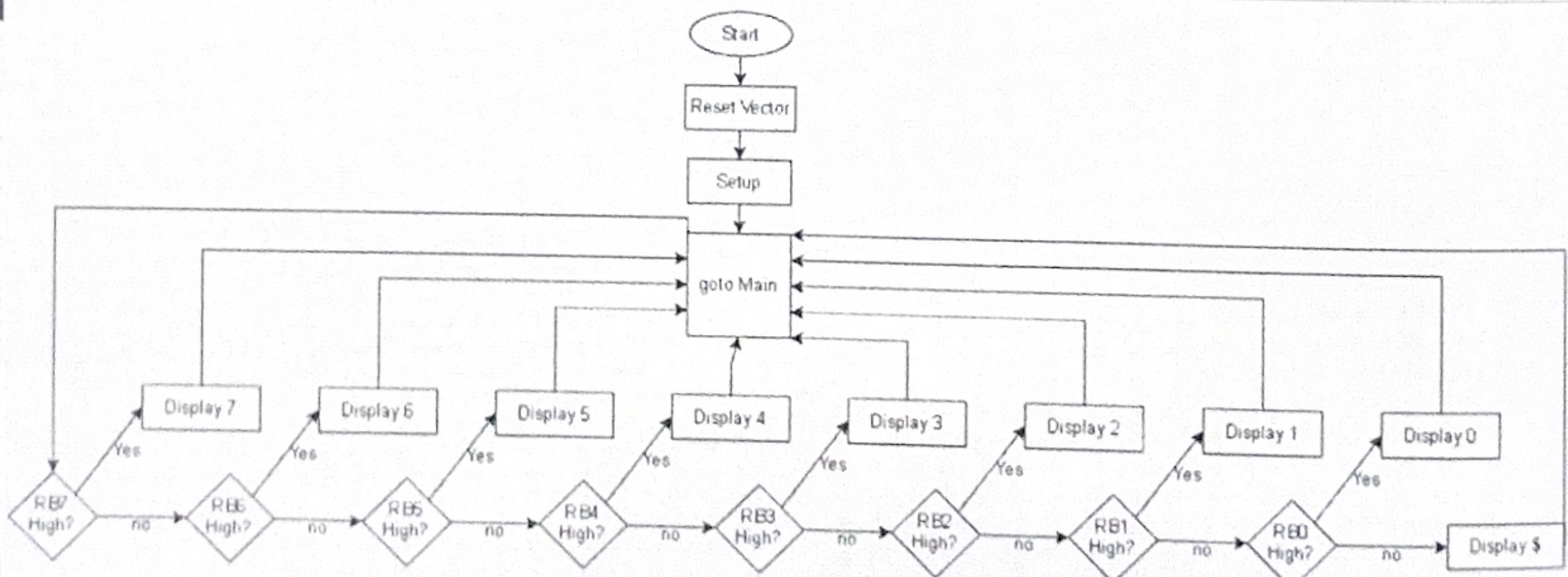
- PORTB input & PORTC output SFR documentation
- Complete DIP switch display schematic
- Loading calculations for all pins & components
- Behavior table
- Priority Selection flow chart
- PIC-As Source

(Full SFR Lists on Pgs 309-312)

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Register on Page
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	122
CCP2CON	—	—	DC2B1	DC2B0	CCP2M3	CCP2M2	CCP2M1	CCP2M0	123
PORTC	RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0	54
PSTRCON	—	—	—	STRSYNC	STRD	STRC	STRB	STRA	144
RCSTA	SPEN	RX9	SREN	CREN	ADDEN	FERR	OERR	RX9D	158
SSPCON	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	177
T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON	81
TRISC	TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0	54

Legend: x = unknown, u = unchanged, — = unimplemented locations read as '0'. Shaded cells are not used by PORTC.

Table 3-1 SFR's Associated with PORT C
PORTB SFR Associations on pg 148



Code Flow chart Figure 3-2

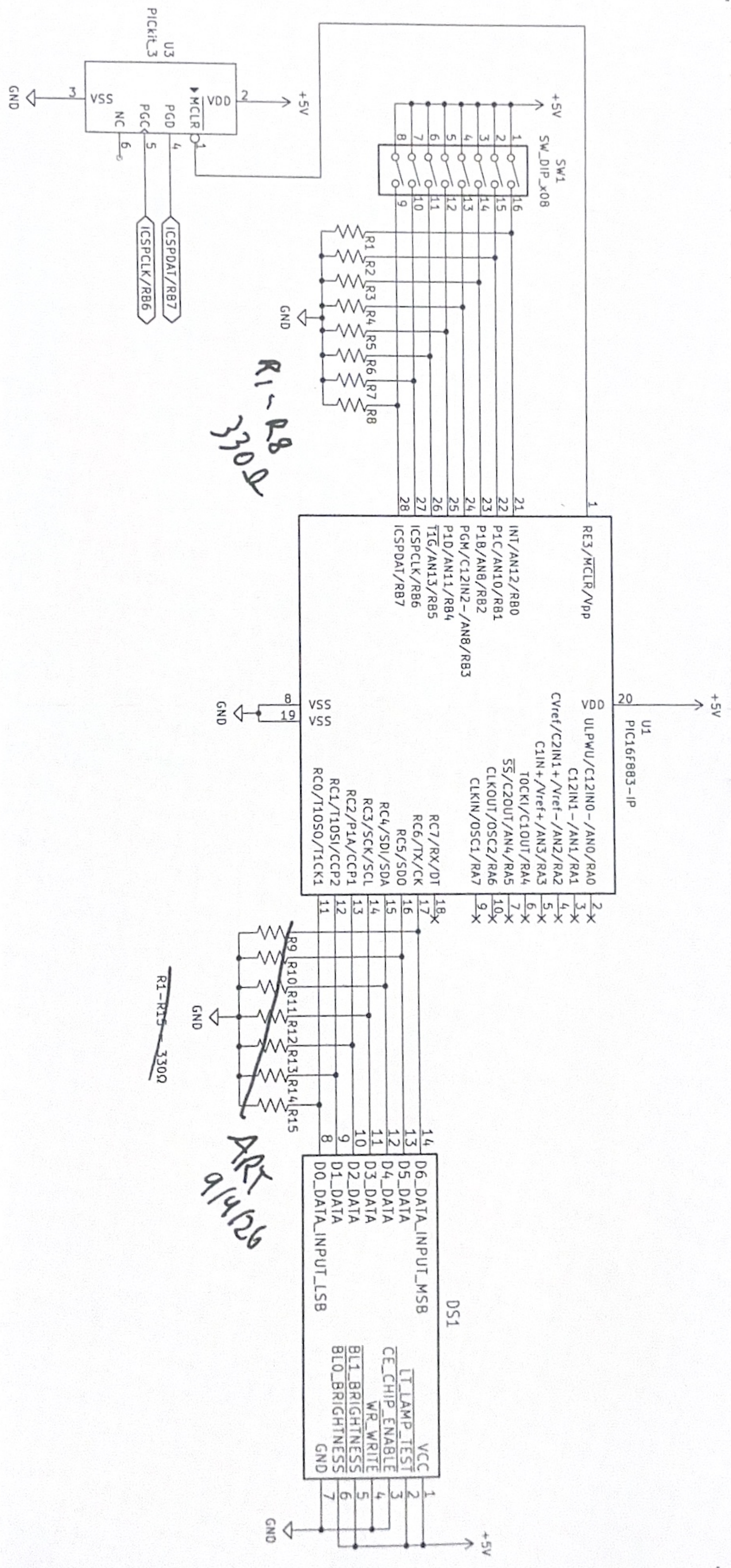


Figure 3-1 PIC16F883 connected to display.

Param No.	Sym.	Characteristic	Min.	Typ†	Max.	Units	Conditions
D030	V _{IL}	Input Low Voltage					
D030A		I/O Port:					
D031		with TTL buffer	V _{SS}	—	0.8	V	4.5V ≤ V _{DD} ≤ 5.5V
D032		with Schmitt Trigger buffer	V _{SS}	—	0.15 V _{DD}	V	2.0V ≤ V _{DD} ≤ 4.5V
D033		MCLR, OSC1 (RC mode) ⁽¹⁾	V _{SS}	—	0.2 V _{DD}	V	2.0V ≤ V _{DD} ≤ 5.5V
D033A		OSC1 (XT and LP modes)	V _{SS}	—	0.3	V	
D033A		OSC1 (HS mode)	V _{SS}	—	0.3 V _{DD}	V	
D040	V _{IH}	Input High Voltage					
D040A		I/O ports:					
D041		with TTL buffer	2.0	—	V _{DD}	V	4.5V ≤ V _{DD} ≤ 5.5V
D042		with Schmitt Trigger buffer	0.25 V _{DD} + 0.8	—	V _{DD}	V	2.0V ≤ V _{DD} ≤ 4.5V
D043		MCLR	0.8 V _{DD}	—	V _{DD}	V	2.0V ≤ V _{DD} ≤ 5.5V
D043A		OSC1 (XT and LP modes)	1.6	—	V _{DD}	V	
D043B		OSC1 (HS mode)	0.7 V _{DD}	—	V _{DD}	V	
D043B		OSC1 (RC mode)	0.9 V _{DD}	—	V _{DD}	V	(Note 1)
D060	I _{IL}	Input Leakage Current ⁽²⁾					
D061		I/O ports	—	± 0.1	± 1	μA	V _{SS} ≤ V _{PIN} ≤ V _{DD} , Pin at high-impedance
D063		MCLR ⁽³⁾	—	± 0.1	± 5	μA	V _{SS} ≤ V _{PIN} ≤ V _{DD}
D063		OSC1	—	± 0.1	± 5	μA	V _{SS} ≤ V _{PIN} ≤ V _{DD} , XT, HS and LP oscillator configuration
D070*	I _{PUR}	PORTB Weak Pull-up Current	50	250	400	μA	V _{DD} = 5.0V, V _{PIN} = V _{SS}
D080	V _{OL}	Output Low Voltage ⁽⁵⁾					
D080		I/O ports	—	—	0.6	V	I _{OL} = 8.5 mA, V _{DD} = 4.5V (Ind.)
D090	V _{OH}	Output High Voltage ⁽⁵⁾					
D090		I/O ports	V _{DD} - 0.7	—	—	V	I _{OH} = -3.0 mA, V _{DD} = 4.5V (Ind.)

Table 3-2 Digital I/O Pin Condition

Absolute Maximum Ratings^(†) for PIC16F883

Ambient temperature under bias	-40° to +125°C
Storage temperature	-65°C to +150°C
Voltage on V _{DD} with respect to V _{SS}	-0.3V to +6.5V
Voltage on MCLR with respect to V _{SS}	-0.3V to +13.5V
Voltage on all other pins with respect to V _{SS}	-0.3V to (V _{DD} + 0.3V)
Total power dissipation ⁽¹⁾	800 mW
Maximum current out of V _{SS} pin	95 mA
Maximum current into V _{DD} pin	95 mA
Input clamp current, I _{IK} (V _I < 0 or V _I > V _{DD})	± 20 mA
Output clamp current, I _{OK} (V _O < 0 or V _O > V _{DD})	± 20 mA
Maximum output current sunk by any I/O pin	25 mA
Maximum output current sourced by any I/O pin	25 mA
Maximum output current sunk by any I/O pin	25 mA
Maximum output current sourced by any I/O pin	25 mA

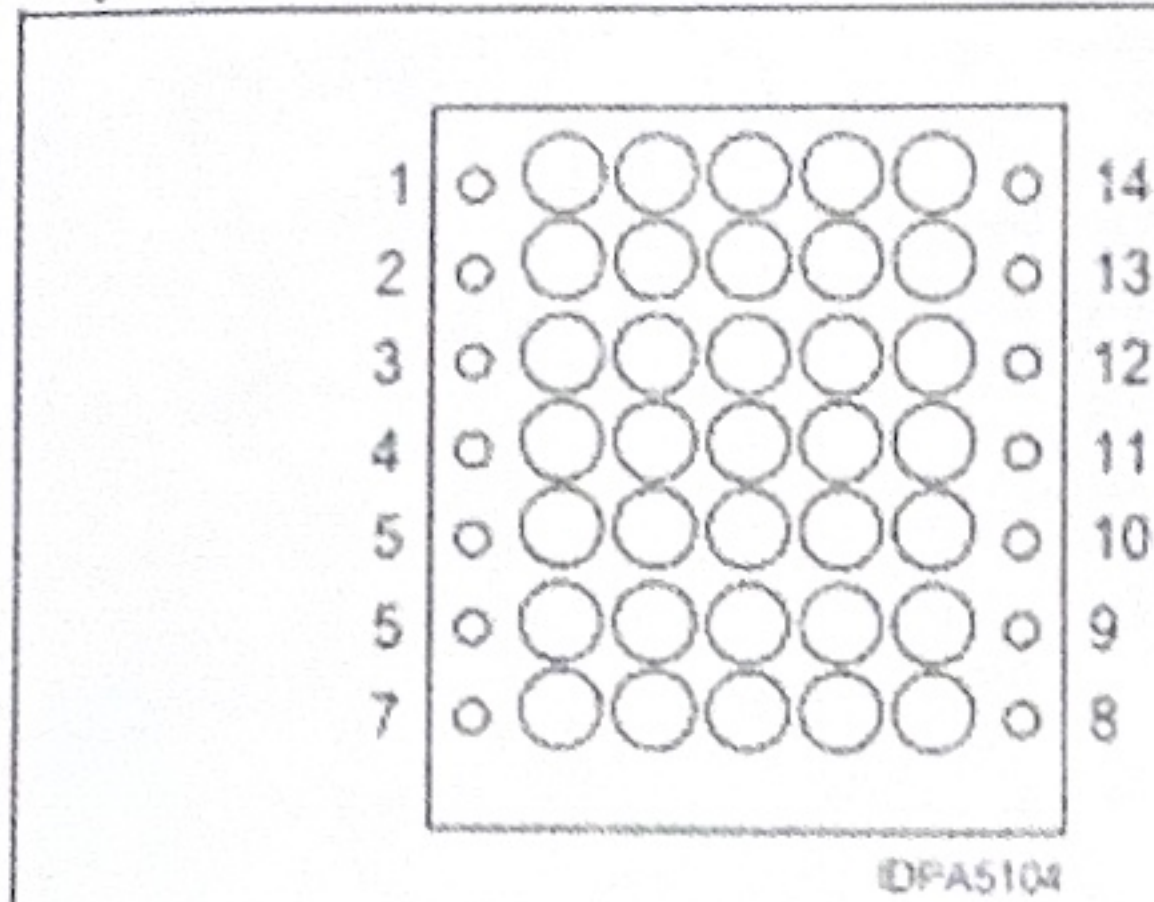
Note 1: Power dissipation is calculated as follows: P_{DIS} = V_{DD} × (I_{DD} - Σ I_{OH}) + Σ ((V_{DD} - V_{OH}) × I_{OH}) + Σ (V_{OL} × I_{OL}).

Worst case calcs

Input Pull down resistor V_{IL} / I_{IL} = 0.8V / 1μA

as long as R₁ - R₈ are below 800K we are safe

Top View — DLO7135, DLG7137



Pin Functions — DLO7135, DLG7137

Pin	Function	Pin	Function
1	V _{CC}	8	D0 Data Input LSB
2	LT Lamp Test	9	D1 Data
3	CE Chip Enable	10	D2 Data
4	WR Write	11	D3 Data
5	BL1 Brightness	12	D4 Data
6	BL0 Brightness	13	D5 Data
7	GND	14	D6 Data Input MSB

ASCII CODE	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	DA	DB	DC	DD	DE	DF
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

THESE CODES DISPLAY BLANK

Table 3-4 character display Hex information Behavior chart

Display Pinout
Figure 3-3
Table 3-3

Data Loading Example

CE	WR	BL0	BL1	LT	Data Input							
					D6	D5	D4	D3	D2	D1	D0	
H	X	H	X	H	X	X	X	X	X	X	X	NC
X	X	L	L	H	X	X	X	X	X	X	X	Blank
X	X	X	X	L	X	X	X	X	X	X	X	Lamp Test
L	L	H	H	H	H	L	L	L	L	L	H	A
L	L	H	H	H	H	H	H	L	L	H	L	r
L	L	H	H	H	L	H	H	L	L	H	H	3
L	L	H	H	H	L	H	L	H	L	H	H	+

X=don't care, NC=no change

Table 3-5 Lamp testing brightness & Enable

DC Characteristics

Parameter	-40°C			+25°C			+85°C			Units	Conditions
	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
I _{CC} (20 dots on)	—	135	—	—	100	140	—	85	—	mA	V _{CC} =5.0 V, BL0=BL1=5.0 V
I _{CC} Blank	—	2.0	—	—	1.5	4.0	—	0.8	—	mA	V _{CC} =WR=5.0 V, BL0=BL1=0 V
I _B (all inputs)	—	—	—	—	50	100	—	—	—	μA	V _{IN} =0.8 V, V _{CC} =5.0 V ± 0.5 V
V _{OL}	2.0	—	—	2.0	—	—	2.0	—	—	V	V _{CC} =5.0 V ± 0.5 V
V _{IL}	—	—	0.8	—	—	0.8	—	—	0.8	V	V _{CC} =5.0 V ± 0.5 V
V _{CC}	4.5	5.0	5.5	4.5	5.0	5.5	4.5	5.0	5.5	V	—

Table 3-6 Digital input specs

In the Lab

1. Verify every individual pin switch
2. Verify the no-switch condition
3. Test multiple switch combinations
4. Confirm that the largest active switch wins
5. Troubleshoot unstable inputs incorrect characters or priority errors.

The only Port C SFRs that are important right now are TRISC to configure it as an output & make sure Port C starts low.

Test7:

```

BTFSS 0x06,7    ;Skip next instruction if RB7 is high
GOTO Test6      ;If RB7 low move to next bit
GOTO Display7   ;If RB7 high display 7

```

each port B pin will be tested in priority order & move immediately to display without testing lower numbers if the pin is tested high. Display code is then sent to port C

Display7:

```

MOVLW 0x37      ;Select the character set for display 7
MOVWF 0x07      ;Send W to PortC
GOTO Main

```

full code in repository on github (pg 147).

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Part 4 - Matrix Keypad Scanning

Goal

Interface a 4x4 matrix keypad, scan the keypad, identify the pressed key & display its assigned value on the DLG7137

Key values & Priority

Key	0-9	A-D	*	#
Value	0x0-0x9	0xA-0xD	0xE	0xF

If multiple keys are pressed, the largest assigned key value must have priority

Before LabDetermine:

- The Keypad row/column arrangement
- PIC Pin assignments
- which lines are driven & which are read during each scan step
- The scan order needed to preserve priority
- all required SFR configuration

Prepare:

- Complete Keypad/Display Schematic
- Loading calculations
- Scan state table
- Flowchart
- PIC-As source

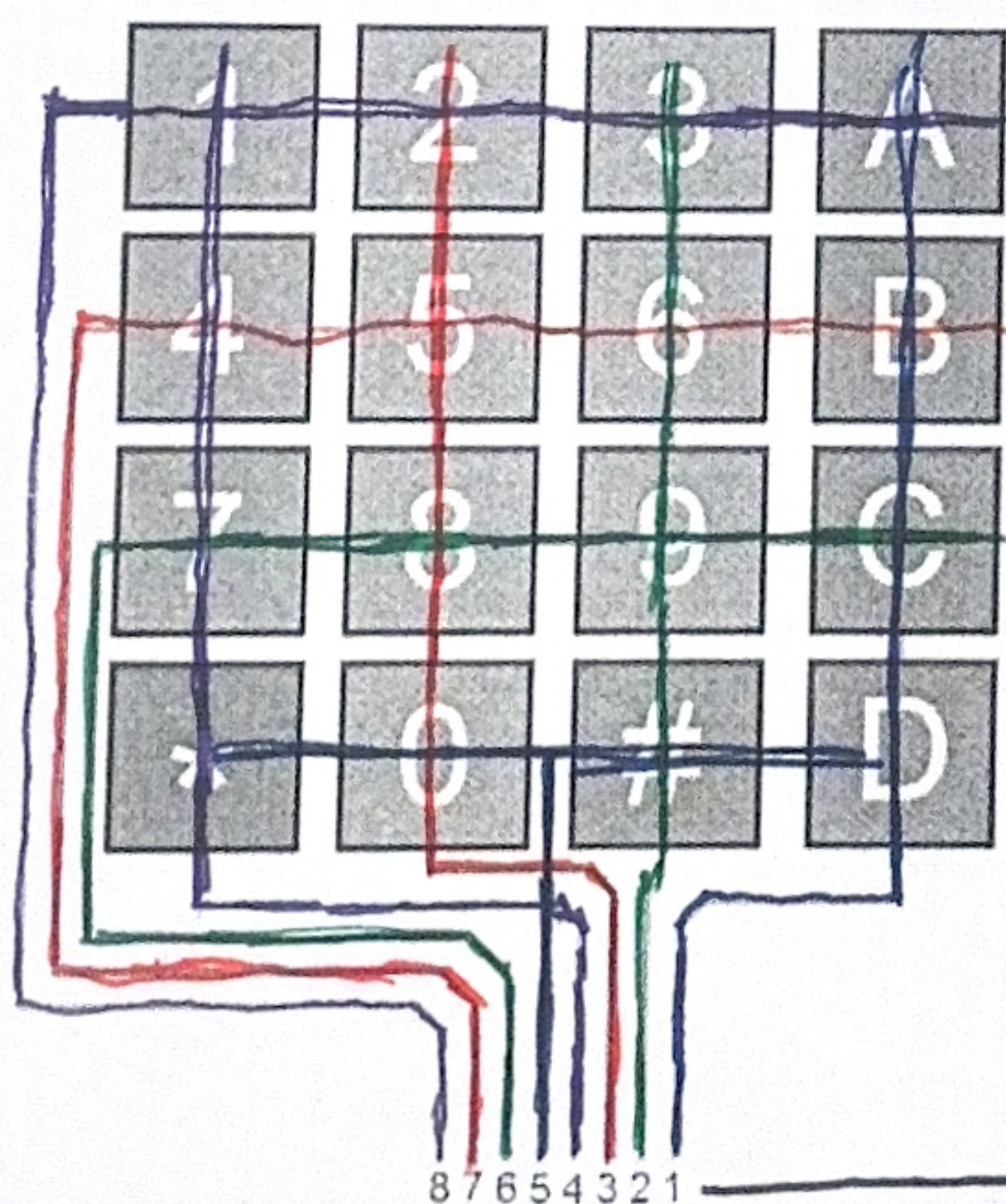


Figure 4-1 Keypad Pinout

Each Column pin is set as inputs permanently while each row is individually made an output, one at a time as to never allow the possibility that two outputs sink/source each other.

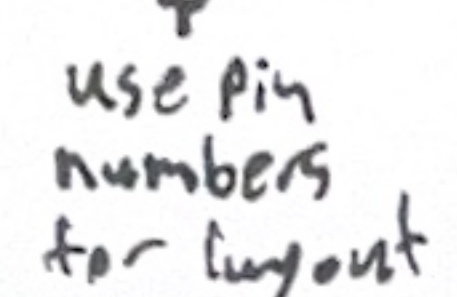
RB0 = 5
RB1 = 6
RB2 = 7
RB3 = 8
RB4 = 4
RB5 = 2
RB6 = 3
RB7 = 4

Priority ->	#:
2,5	
4,5	
D:1,5	
C:1,6	
B:1,7	
A:1,8	
9:2,6	
8:3,6	
7:4,6	
6:2,7	
5:3,7	
4:4,7	
3:2,8	
2:3,8	
1:4,8	
0:3,5	

Figure 4-2
Priority
Order

Code on github repo (pg 147)

full SFR documentation Pg 309-312



Setun:

```

; add device setup here
BSF 0x03, 6 ;
BSF 0x03, 5 ; Set bank 3 active
CLRf 0x89 ; ANSELH set low to make portB digital

BSF 0x03, 6 ;
BCF 0x03, 5 ; Set bank 2 active
BSF 0x09, 1 ; Turn T1G-not function off

BCF 0x03, 6 ;
BSF 0x03, 5 ; Set bank 1 active
CLRf 0x87 ; TrisC sets all portC pins as outputs
MOVLW 0xFF ; Put all high literal into W
MOVWF 0x86 ; Put all highs on TrisB to configure PortB as inputs
MOVLW 0xFF ; Put all high literal into W
MOVWF 0x95 ; WPUB
BCF 0x81, 7 ; OPTION_REG

BCF 0x03, 6 ;
BCF 0x03, 5 ; Set bank 0 active
CLRf 0x06 ; Clear portB
CLRf 0x07 ; Clear portC
; IOCB portB interrupt-on-change disabled default

```

```

L goto Main

```

Worst Case calcs

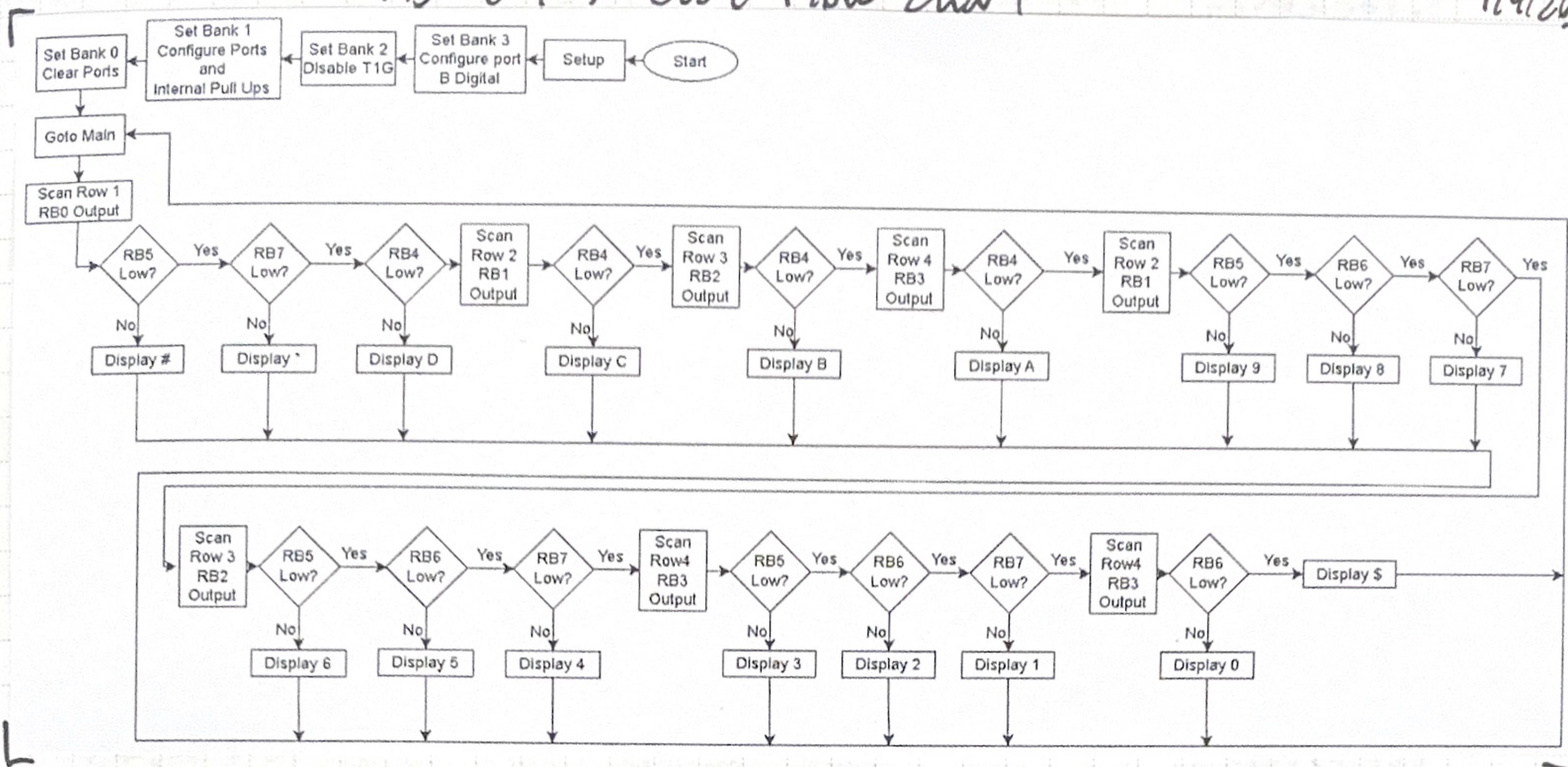
Input pull up resistors $V_{IH} / I_{inpull} = 2V / 250\mu A$

as long as $R_1 - R_2$ are below R_K we are fine

Experiment 2 PIC16F883 I/O Familiarization

Figure 4-5 Code Flow chart

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In the Lab

1. Verify all 16 Keys individually
2. Confirm the correct assigned value is displayed
3. Observe at least one complete scan sequence with an O-Scope or logic analyzer
4. Compare the measured scan with your scan table
5. Measure the scan period

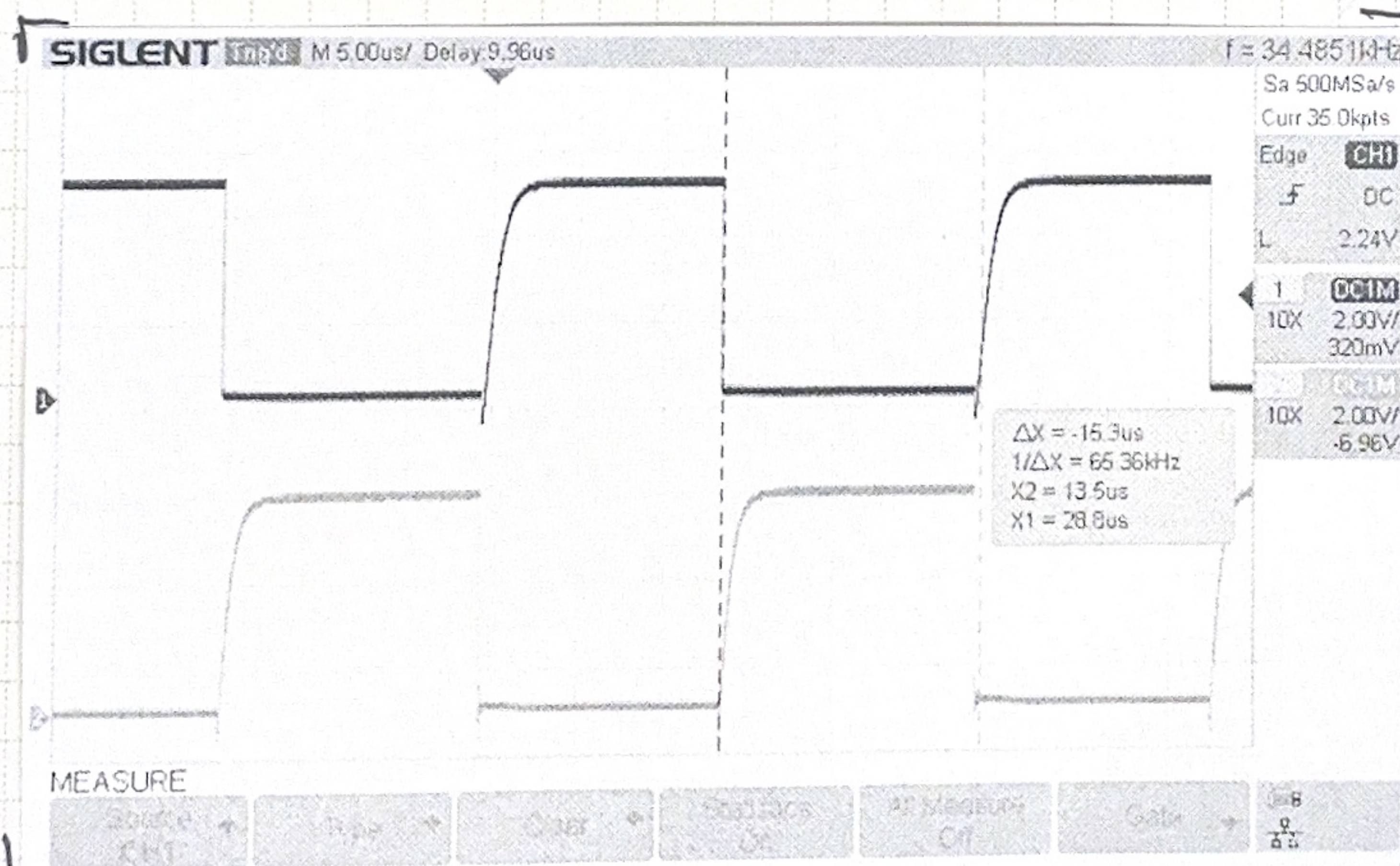


Figure 4-6 measured Scan period @ 30x5

6. Test multiple simultaneous key presses after protection is ensured
7. Confirm priority order

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8. Document Trouble shooting changes.

The only trouble shooting changes that were made were to adjust when the Row outputs were initiated by placing them INSIDE Test routines. This paired with adding simple bank switching code before & after the Row tests was all the trouble shooting changes.

Conclusion

This lab was focused on the introduction of the PIC16F883 Micro controller. It had us explore MPLab, Set up base code Configuration for easy startup on future Programs, Set up repository connections, & Linker code. We then explored how to use Instruction code to change special function registers & configure a single port to increment infinitely. Then we used similar code in part 3 to cause an LED display to act as a priority encoder display for numbers 0-7 on a DIP-switch. We then applied similar logic to a 4x4 Matrix Key pad to display its button presses with priority.

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